

Agustín Roca

Software Engineer

PhD Student

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📍 Buenos Aires, Argentina

EDUCATION

Applied Sciences PhD student - Universidad de San Andrés (UdeSA)

10/2024 - 12/2029

Software Engineer - Instituto Tecnológico de Buenos Aires (ITBA)

03/2017 - 12/2022

WORK EXPERIENCE

Research Engineer - AI and Robotics Laboratory (Universidad de San Andrés)

Buenos Aires, Argentina

- Integrate and synchronize LiDAR, cameras, IMU and GPS to an autonomous car system
- Implement AI computer vision system for deer detection in UAV aerial imagery

02/2023 - Present

Head Teaching Assistant - Universidad de San Andrés

Buenos Aires, Argentina

- Computational thinking: Introductory course to programming in Python and C
- Organize and give practical classes, including Hands-on projects for 30+ students
- Coordinate practical classes with three other simultaneous courses

02/2023 - Present

Software Engineer Intern - J.P. Morgan Chase

Buenos Aires, Argentina

- Fully develop and test software to create efficient solutions to company workflows
- Migration from Python 2 to Python 3
- Communicate with people around the world working in different areas of the company

11/2021 - 04/2022

PUBLICATIONS

- A. Roca, G. Castro, G. Torre, L. Colombo, I. Mas, J. Pereira & J.I. Giribet

"Detection of Endangered Deer Species Using UAV Imagery: A Comparative Study Between Efficient Deep Learning Approaches"

ICUAS 2025 (Accepted for publication)

- A. Roca, G. Torre, J. Giribet, G. Castro, L. Colombo, I. Mas & J. Pereira

"Efficient Endangered Deer Species Monitoring with UAV Aerial Imagery and Deep Learning"

IEEE ARGENCON 2024. <https://ieeexplore.ieee.org/abstract/document/10735858>

- J. Giribet, I. Mas, A. Roca, G. Marzik, G. Torre, C. Rosito & G. Castro

"Base de Datos para Conducción Autónoma. Sensores y Sincronización"

Presented at Jornadas Argentinas de Robótica 2024

Dataset published at CONICET institutional repository: <https://ri.conicet.gov.ar/handle/11336/242774>

- A. Roca & N. I. Britos

"Controlling Face's Frame generation in StyleGAN's latent space operations"

ITBA. Engineering Thesis. 2022. <https://ri.itba.edu.ar/handle/123456789/4116>

CONFERENCES

- **KHIPU 2025** - Latin American Meeting in Artificial Intelligence (Poster presented)
- **UdeSA's SCIAA 2024** - Applied Artificial Intelligence Scientific Symposium (Poster presented)
- **JAR 2024** - Argentinean Robotics Conference (Oral presentation given)

CERTIFICATES

- **Getting Started with Deep Learning** - NVIDIA - 03/2025
- **Create and Manage Cloud Resources** - Google - 03/2022
- **Google Cloud Essentials** - Google - 03/2022

SKILLS

AI ML Python Computer Vision Research Teaching ROS Videogames VueJS SQL VR Bioinformatics UX

Unity Team working Data Visualization C Java Android development Web development Agile development MS Office

LANGUAGES

Spanish - Native Proficiency

English - Bilingual Proficiency - IELTS score: 7.0/9.0